

7. The art of matryoshka involves making a set of wooden dolls of decreasing size that are placed one inside another.



When one doll is opened up, a smaller one can be found inside, until eventually the last and smallest doll is reached. The smallest doll is a solid piece of wood.



A student states 'Particle physics is just like matryoshka dolls. Look inside a particle and you will always find something similar, just smaller.'

Using your knowledge of physics, comment on this statement.

3





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7. (continued)



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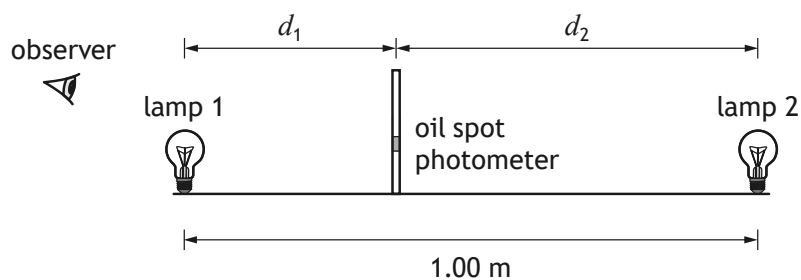


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8. A group of students is conducting an experiment to investigate the relationship between the power output of two small filament lamps and the relative brightness of each lamp.

The relative brightness can be determined using an oil spot photometer.

The experiment is set up as shown.



The oil spot photometer is a piece of card with an oil spot on it, which is used to compare the brightness of two light sources on either side of the photometer. When the relative brightness of the light on either side of the photometer is equal, no spot is observed.

